

# GenAI for Patient-Centric Healthcare

Practical, responsible workflows for doctors and pharma professionals

60 MIN THEORY

90 MIN HANDS-ON

SYNTHETIC DATA

Four pillars • medical-specific demos • qualified human decisions

Patient engagement

Clinical workflow

Pharma & research

Responsible AI

OPENING

# The learning contract

Useful output is not the goal. Verified, accountable workflow improvement is.

1

## Use approved data

Synthetic, public, or institution-approved de-identified inputs only.

2

## Bound the task

Draft, extract, structure, compare, or explain—never silently decide.

3

## Verify the source

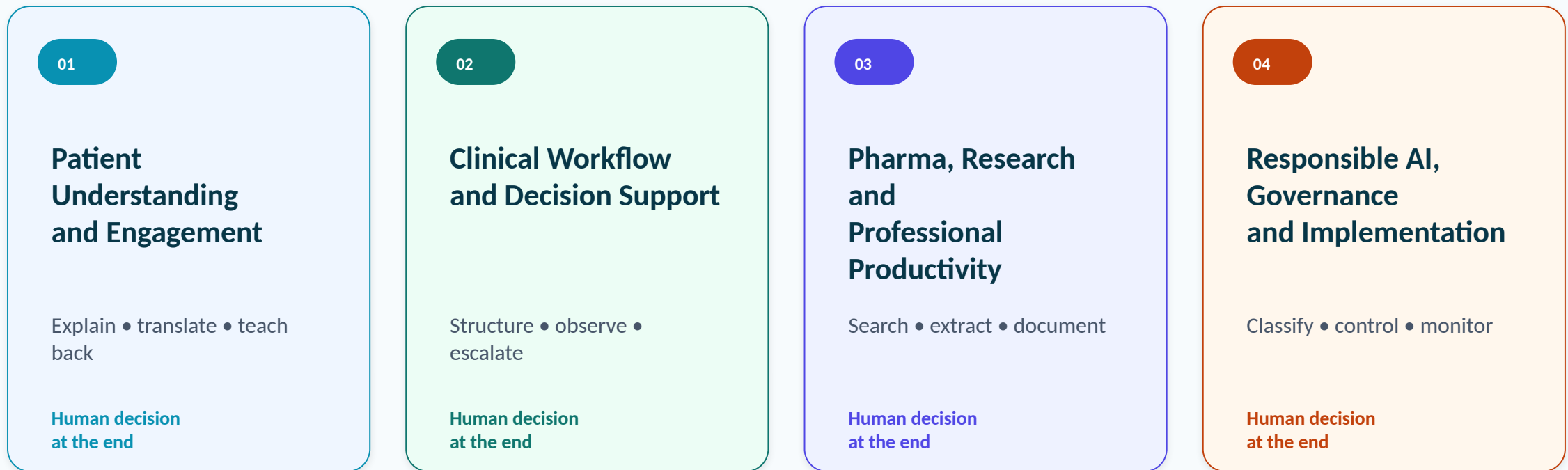
Check every consequential fact, citation, ambiguity, and omission.

4

## Keep human authority

Record accept, edit, reject, or escalate—and who owns it.

# Four pillars—from patient need to responsible implementation



PILLAR 1

# Prescription extraction: separate reading from explaining

A synthetic image, an exact source table, and a clinician/pharmacist decision.

**ALKEM AI MASTERCLASS**  
SYNTHETIC PRESCRIPTION • NOT FOR CLINICAL USE

Dr. Ananya Kulkarni • Demonstration Clinic  
Prescription ID: RX-SYN-001

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Synthetic patient: Aarav Mehta

**Rx Metformin 500 mg**  
 One tablet with evening meal  
 Duration: 30 days

Follow-up / warning from source

Review in 4 weeks

SIGNATURE

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This image and every name/value on it are fully synthetic.  
 Use only to practise extraction, explanation, and human verification.

1

## Extract

Transcribe legible fields exactly.

2

## Flag

Mark ambiguity [UNREADABLE]; never guess.

3

## Explain

Use plain language without adding advice.

4

## Verify

Compare field-by-field; record human decision.

## Never let the model

- reconstruct unclear text
- infer the indication
- prescribe or substitute
- assess interactions autonomously
- overrule the source prescription

**Owner: clinician or pharmacist**

PILLAR 1

# From discharge summary to patient action plan

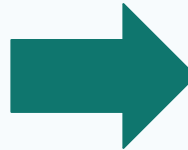
Preserve the medicine, date, warning, and follow-up—change only clarity.

## CLINICIAN SOURCE

**“Continue medication as listed. Review as advised. Return if symptoms worsen.”**

What is missing for a patient?

- exact action
- urgency threshold
- timing
- comprehension check



## PATIENT ACTION PLAN

1. What happened—in plain words
2. Exact medicine/action schedule
3. Warning signs and urgent route
4. Follow-up: who, when, where
5. Three teach-back questions

**Bilingual drafts require fluent clinical review.**

# Medical images: demonstrate the boundary, not a diagnosis

General-purpose image models are not a substitute for qualified interpretation or validated medical-device workflows.

FULLY SYNTHETIC • NO GROUND TRUTH



## Practise

- ✓ image adequacy
- ✓ neutral visible observations
- ✓ cannot-assess items
- ✓ uncertainty
- ✓ human escalation

## Do not claim

- ✗ diagnosis
- ✗ emergency clearance
- ✗ treatment advice
- ✗ “normal” or “safe” verdict
- ✗ validated performance

# Clinical reasoning: use AI to reduce omission, not own the decision

COMMON

**Common / likely**

Keep prevalence and presentation in view.

CAN'T MISS

**Cannot-miss**

Separate severity from probability.

MISSING

**History / exam gaps**

Make absent information visible.

ESCALATE

**Urgency triggers**

Define when human action cannot wait.

**Always ask: What evidence supports this? What opposes it? What would change urgency?**

# Two lanes: safety intake and evidence verification

Both gain speed from structure. Neither delegates qualified judgement.

## PHARMACOVIGILANCE

### AI may help

- structure chronology
- screen seriousness fields
- identify missing data
- draft neutral follow-up

**Never decide causality, expectedness, coding, or reportability.**

## EVIDENCE

### AI may help

- form PICO/PECO
- expand search concepts
- extract supplied sources
- map evidence gaps

**Never invent a citation.  
Open and verify every source.**

**Human-owned: safety decisions • screening log • source checks • interpretation • disclosure**



## PILLAR 4

# Risk follows data × impact × autonomy × reach

Use the highest-risk dimension to set minimum controls and a stop condition.

DATA

Synthetic → identifiable / highly sensitive

LOWER

HIGHER

IMPACT

Communication → influences care

LOWER

HIGHER

AUTONOMY

Draft → default → autonomous action

LOWER

HIGHER

REACH

Sandbox → patient / population scale

LOWER

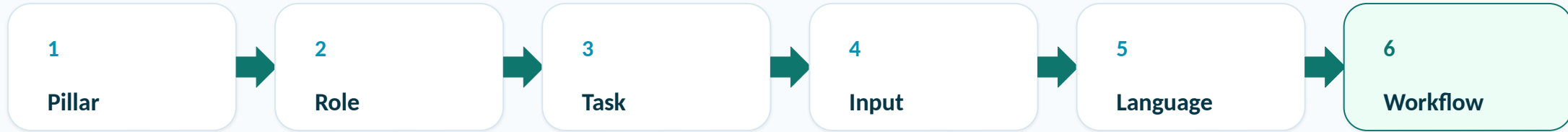
HIGHER

## Minimum decision record

- accountable owner
- allowed data
- human review
- monitoring metric
- incident route
- stop / rollback rule

# Choose the workflow first; let supported options cascade

The app offers only intersections backed by a complete prompt, dataset, output, verification, and reviewer.

**DATA FIT**

Can this account and workflow process the intended data?

**CAPABILITY**

Does the tool support the input and output—and with what limitations?

**EVIDENCE**

Can claims and citations be opened, traced, and checked?

**HUMAN FIT**

Who reviews, what gets escalated, and what stops the workflow?

# 90 minutes: six medical and pharma exercises

For every lab: 3 min orient • 6 min run • 4 min verify • 2 min decide.

01

## Prescription extraction

Exactness • ambiguity • explain

02

## Image observation

Adequacy • boundary • escalate

03

## Discharge + language

Meaning • teach-back • review

04

## Differential + red flags

Omission • bias • urgency

05

## PV intake

Neutrality • missingness • route

06

## Evidence verification

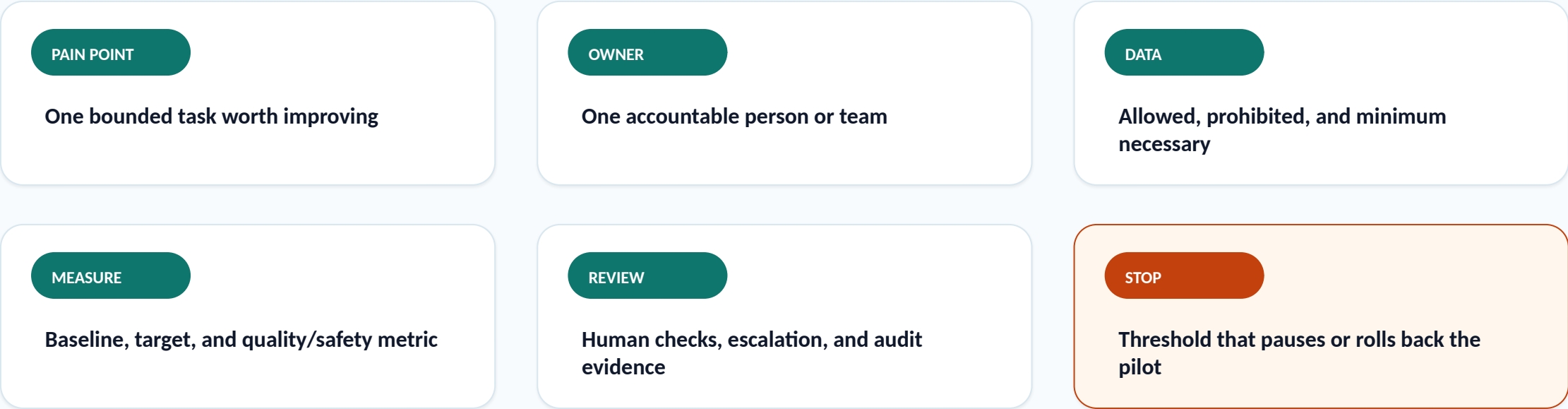
Search log • source checks

Decision record for every output: ACCEPT • EDIT • REJECT • ESCALATE

CLOSE

# Your 30-day responsible pilot

Start with a reversible workflow, a measurable outcome, and a stop rule.



The model drafts. The source constrains. The qualified human decides.

TAKE THE WORKBOOK + APP + VERIFIED TOOL MATRIX